Incident report analysis

Summary	The organization experienced a DDOS Distributed Denial of Service attack recently that compromised the internal network for 2 hours until resolution. It was discovered that during the attack the organizations network services stopped responding due to an incoming flood of ICMP Internet Control Message Protocol pings into the network through an unconfigured firewall. Because of this normal internal network traffic could not access any network resources. The cybersecurity team investigated this and discovered that a malicious actor had sent a flood of ICMP pings into the company's network through an unconfigured firewall. Because of this the company's network was overwhelmed through this DDOS attack. In response to this the incident management team blocked incoming ICMP packets, stopped all non critical network services offline, and restored critical network services.
Identify	According to the cybersecurity team this was a DDOS Distributed Denial of Service Attack. The type of DDOS attack this was is an ICMP Internet Control Message Protocol flood attack. This is when ICMP packets are sent an excessive amount of times from different IP addresses unless IP spoofing is being used, which is done to disrupt or shut down a network and its services. The system affected was the internal network of the organization and network services.

Protect	To protect and respond to this incident some network hardening techniques needed to be added and implemented. The network security team implemented a new firewall rule to limit the rate of incoming ICMP packets and IDS/IPS's which are Intrusion Detection Systems/Intrusion Prevention Systems to filter out some ICMP traffic based on suspicious characteristics.
Detect	The security team implemented source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets and network monitoring software to detect abnormal traffic patterns. To detect abnormal network traffic in the future the team can use network log analysis tools like SIEM System Information and Event Management tools to monitor all network traffic on the network and firewall maintenance to ensure that the firewall is configured properly and updated for events like this. Additionally, unused hardware and software applications should be disposed of and network access privileges to allow, limit, or block privileges from certain users, IP addresses, mac addresses etc.
Respond	For future security events the team will isolate all infected systems to prevent further disruption. The team will attempt to restore any critical systems and network services disrupted by the event. The incident management team did block incoming ICMP packets and stop all non critical network services to reduce internal network traffic during the incident to restore critical systems and network services. The team will analyze network logs to check for any

	unusual or suspicious activity. The team will report all incidents to upper management and legal authorities, if applicable.
Recover	To recover from this DDoS attack by ICMP flooding, access to network services need to be restored to a normal functioning state. In the future, external ICMP flood attacks can be blocked at the firewall. Then, all non-critical network services should be stopped to reduce internal network traffic. Next, critical network services should be restored first. Finally, once the flood of ICMP packets have timed out, all non-critical network systems and services can be brought back online.

Reflections/Notes: